

# Reviewer Pack — Coxeter Group Order of Polynomial Systems vs. Resolution Pro...

Entry 75acd35fcc70 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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## Coxeter Group Order of Polynomial Systems vs. Resolution Proof Length

Entry ID: 75acd35fcc70 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-24 20:05:32 UTC

### 1. Conjecture

**Field A** (mathematical branch): Coxeter Groups and Hecke Algebras **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity

#### Statement:

[‘For every polynomial system  $f$  over the field  $F_2$ , let  $\rho(f)$  be the order of the unique irreducible representation associated with  $f$  in a Coxeter group. Then, for all instances of Tseitin formulas,  $\rho(f) = \Theta(\log n)$ , where  $n$  is the number of variables in the formula, and the resolution proof length is upper-bounded by  $2^{\rho(f)}$ .’, ‘Equivalently, the length of any resolution proof of a Tseitin formula is polynomially related to the order of the associated irreducible representation in a Coxeter group.’]

#### Rationale (proposer’s reasoning):

[‘Coxeter groups and Hecke algebras have been used to study representations of algebraic objects. By associating irreducible representations with polynomials, we can potentially uncover structural properties of these representations that are relevant to the complexity of resolution proofs.’, ‘The conjecture aims to link polynomial systems, which are fundamental in many computational problems, with Coxeter groups, a tool from representation theory. If true, this would suggest that certain algebraic structures underlying the problem could be exploited to design more efficient algorithms.’]

**Taxonomy category:** GCT\_DET\_PERM (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix):** f475806c2efb690d

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

#### Acceptance criterion:

The conjecture is supported if the mean order of irreducible representations  $\rho(f)$  for Tseitin formulas of size  $n \leq 40$  is within a factor of 1.5 of  $\log(n)$ , and the resolution proof length is less than or equal to  $2^{(1.25 * \rho(f))}$ . The conjecture is falsified if either condition is not met for at least one seed.

### 3. Barrier filter (F1)

**Final:** PASS

| Barrier        | Verdict | Confidence | LLM <sub>1</sub> | LLM <sub>2</sub> |
|----------------|---------|------------|------------------|------------------|
| RELATIVIZATION | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| NATURAL_PROOFS | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| ALGEBRIZATION  | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |
| KARP_LIPTON    | SAFE    | 0.50       | UNCERTAIN        | UNCERTAIN        |

### 4. Novelty audit

**Verdict:** NOVEL against 2 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - "irreducible representation" AND "Coxeter group" AND "polynomial system" - "resolution proof complexity" AND "Coxeter group" AND "Tseitin formulas" - "Hecke algebra" AND "resolution proof length" AND "irreducible representation order"

**Top relevant hits considered:** - [s2:10.54254/2753-8818/43/20241036] Representation of coxeter group and orthogonal group - [s2:10.1017/CBO9780511623646.008] Reflection groups and Coxeter groups: Hecke algebras and Kazhdan-Lusztig polynomials

### 5. Empirical test harness

**Multi-seed protocol:** 5 seeds (11, 23, 37, 53, 71)

**Sandbox:** isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

#### 5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_tseitin_formula(n):
        variables = [f'x{i}' for i in range(n)]
        clauses = []
        for var in variables:
            clauses.append([var])
        for i in range(1, n):
            new_var = f'y{i}'
            clauses.append([new_var, f'~{variables[i-1]}'])
            clauses.append([new_var, f'{variables[i]}'])
            clauses.append([f'~{new_var}', f'~{variables[i-1]}', f'~{variables[i]}'])
        return variables, clauses

    def calculate_rho(f):
        # Placeholder for actual calculation of rho(f)
        # This is a dummy implementation
        return random.randint(1, 5)

    def construct_resolution_proof(clauses):
        proof_length = len(clauses) * 2
```

```

        return proof_length

n_values = [5, 10, 15, 20, 30, 40]
total_rho = 0
total_proof_length = 0
instances_tested = 0

for n in n_values:
    variables, clauses = generate_tseitin_formula(n)
    rho_f = calculate_rho(f"polynomial_system_{n}")
    proof_length = construct_resolution_proof(clauses)

    if rho_f <= 0 or proof_length <= 0:
        continue

    total_rho += rho_f
    total_proof_length += proof_length
    instances_tested += 1

if instances_tested == 0:
    return {
        "metric_name": "rho_and_proof_length",
        "metric_value": None,
        "instances_tested": instances_tested,
        "conjecture_holds": False,
        "counterexample": "No valid instances generated"
    }

mean_rho = total_rho / instances_tested
mean_proof_length = total_proof_length / instances_tested

conjecture_holds = (mean_rho <= 1.5 * math.log(instances_tested, 2)) and (mean_proof_length <= 1.5 * math.log(instances_tested, 2))

return {
    "metric_name": "rho_and_proof_length",
    "metric_value": mean_rho,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": "" if conjecture_holds else f"Mean rho: {mean_rho}, Mean proof length: {mean_proof_length}"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    if all(result["conjecture_holds"] for result in results):
        mean_rho = sum(result["metric_value"] for result in results) / len(results)
        support_fraction = 1.0
        print(f"RESULT: SUPPORTED mean={mean_rho} std=0.0 support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):

```

```

    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\", first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=unknown")

```

## 6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

## 7. Test stdout (last 2KB)

```

.6666666666666665, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 2.3333333333333335, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 3.0, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 2.8333333333333335, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 2.8333333333333335, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 3.1666666666666665, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 3.5, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 3.0, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 3.3333333333333335, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.
TRIAL: {'metric_name': 'rho_and_proof_length', 'metric_value': 2.8333333333333335, 'instances_tested': 6, 'conjecture_holds': False, 'counterexample': 'Mean rho: 2.

--STDERR--
Traceback (most recent call last):
  File "/home/lud

```

## 8. Critic adversarial review

**Critic verdict:** CHALLENGE

**Critic reasoning:**

skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict:** INCONCLUSIVE

**Reasoning:**

Safety rail: critic\_challenge\_falsified | original: The mean order of irreducible representations  $\rho(f)$  for Tseitin formulas does not meet the expected range within a factor of 1.5 of  $\log(n)$ , and the res | next: Investigate the counterexamples to understand why the mean  $\rho(f)$  is outside the expected range and if there are patterns in the data that could lead to a more accurate model or explanation.

## 11. Audit log (LLM calls)

**Total LLM calls:** 9

| # | Phase           | Provider      | Model            | Tokens in | out | Latency (ms) |
|---|-----------------|---------------|------------------|-----------|-----|--------------|
| 1 | propose         | ollama_remote | glm4:latest      | 0         | 0   | 12209        |
| 2 | preregistration | ollama_remote | glm4:latest      | 0         | 0   | 6567         |
| 3 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 4784         |
| 4 | novelty         | ollama_remote | glm4:latest      | 0         | 0   | 5620         |
| 5 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 18048        |
| 6 | test_gen        | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 8099         |

| # | Phase    | Provider      | Model            | Tokens in | out | Latency (ms) |
|---|----------|---------------|------------------|-----------|-----|--------------|
| 7 | test_gen | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 11452        |
| 8 | test_gen | ollama_remote | qwen2.5-coder:7b | 0         | 0   | 10684        |
| 9 | judge    | ollama_remote | glm4:latest      | 0         | 0   | 15239        |

**Totals:** 0 input tokens, 0 output tokens, ~\$0.0000 cost, 92702 ms total latency. Provider mix: {'ollama\_remote': 9}

*(full prompt+response transcripts available in research/audit/75acd35fcc70.jsonl)*

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/75acd35fcc70.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file:** research/pvsnp\_sandbox/test\_\*.py (per-cycle)

**Replay tarball:** research/replay/75acd35fcc70.tar.gz (if generated)